

Sijie RUAN (阮思捷)

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SHORT BIO

Dr. Sijie RUAN is currently an Assistant Professor and Associate Research Fellow of Beijing Institute of Technology (BIT), and a Deputy Director of the Application Innovation Center of Commercial Technology, Ministry of Commerce. His research interests include spatio-temporal data mining, and GeoAI. He has authored more than 60 referred publications at prestigious international conferences and journals (ICML, NeurIPS, AAAI, TKDE, KDD, ICDE). He was awarded Best Paper / Best Demo Paper Award at international conferences for three times. He received Science and Technology Progress Award (First Prize) of Chinese Institute of Command and Control, Doctoral Dissertation Incentive Program of Chinese Institute of Electronics, and Outstanding Ph.D. Thesis Award of ACM SIGSPATIAL CHN Chapter.

WORKING EXPERIENCE

- School of Computer Science and Technology, Beijing Institute of Technology** Beijing, China
Assistant Professor and Associate Research Fellow Jul. 2022 – Present
- JD Intelligent Cities Research, JD.com** Beijing, China
Research Intern Jan. 2018 – Jun. 2021
- School of Computer Science and Technology, Xidian University** Xi'an, Shaanxi, China
Teaching Assistant Sept. 2017 – Nov. 2017
- Urban Computing Group, Microsoft Research Asia** Beijing, China
Research Intern Jul. 2016 – Aug. 2017

EDUCATION EXPERIENCE

- School of Computer Science and Technology, Xidian University** Xi'an, Shaanxi, China
Ph.D. in Computer Science; Advisor: Prof. Yu Zheng (IEEE Fellow) Sept. 2017 – Jun. 2022
- School of Computer Sci. and Eng., Nanyang Technological University** Singapore
Visiting Ph.D. Student; Advisor: Assoc. Prof. Cheng Long Sept. 2019 – Apr. 2022
- School of Computer Science and Technology, Xidian University** Xi'an, Shaanxi, China
Bachelor of Engineering in Computer Science; Excellent Class; Rank: 1/29 Aug. 2013 – Jul. 2017

RESEARCH GRANTS

- National Natural Science Foundation of China**, Research on Methods for Fine-grained Geographical Information Collecting and Urban Space Modeling Driven by Spatio-temporal Big Data (62306033), ¥300K, 2024.01-2026.12, **PI**.
- Guizhou Provincial Key Technology R&D Program**, Research on Key Generic Technologies for Intelligent Data Governance (PA[2025]002), ¥2.00M, 2024.04-2027.03, **PI**.
- National Key Research and Development Program of China**, Development on Model Integration Tools, ¥1.10M, 2023.11-2026.10, **Co-I**.
- National Key Research and Development Program of China**, Research on Technologies of Risk Active Sensing and Prediction for Major Infectious Disease, ¥350K, 2023.12-2026.11, **Co-I**.

PROFESSIONAL APPOINTMENTS

- Deputy Director:** Application Innovation Center of Commercial Technology, Ministry of Commerce
- Senior Area Chair:** KDD (2026)
- Session Chair:** ICDE (2024), UrbComp (2021)
- Publicity Chair:** UrbComp (2024, 2025)
- Executive Committee Member:** CCF Expert Committee on Big Data, ACM SIGSPATIAL CHN Chapter
- Youth Committee Member:** Public Health Surveillance Committee of Chinese Preventive Medicine Association
- Journal Reviewer:** TKDE, TMC, INS, IMWUT, TITS, TGIS, CJE, TIST, TBD, TDS
- Program Committee:** KDD (2023-2026), NeurIPS (2024-2025), ICLR (2025-2026), WWW (2025), AAAI (2021, 2026), CIKM (2022-2025)

HONORS AND AWARDS

- [H1] **The 1st Prize in Science and Technology Progress Award:** Chinese Institute of Command and Control, 2023
- [H2] **Doctoral Dissertation Incentive Program:** Chinese Institute of Electronics, 2024
- [H3] **Outstanding Ph.D. Thesis Award:** ACM SIGSPATIAL CHN Chapter, (1 in total), 2022
- [H4] **Development Award Fund for Excellent Young Scholars:** Beijing Institute of Technology (6 in total), 2025
- [H5] **The 1st Place Award of Travel Planning Challenge:** IJCAI, 2025
- [H6] **Best Paper Award:** ACM CIKM Workshop Frontiers in Graph Machine Learning for the Large Model Era, 2025
- [H7] **Best Demo Paper Award:** ACM SIGSPATIAL, 1/27, 2021
- [H8] **Best Demo Paper Runners-up:** ACM SIGSPATIAL, 2017

BOOKS

- [B1] Translated by Xin Zou, **Sijie Ruan**, Zhiyi Liu, Shuliang Wang. *Deep Learning: Foundations and Concepts (in Chineses)*. Posts & Telecommunications Press.

SELECTED PUBLICATIONS († INDICATES CORRESPONDING AUTHORS, * INDICATES CO-FIRST AUTHORS)

- [P1] Xukai Zhang, Shuliang Wang, Guangyin Jin, Ziqiang Yuan, Hanning Yuan†, **Sijie Ruan**†. *Eulerian Neural Network Informed by Chemical Transport for Air Quality Forecasting*. **NeurIPS’25**.
- [P2] Shuliang Wang, Song Tang, **Sijie Ruan**†, Cheng Long, Yuxuan Liang, Qi Li, Ziqiang Yuan, Jie Bao, Yu Zheng. *Urban Sensing for Multi-Destination Workers via Deep Reinforcement Learning*. **ICDE’24**.
- [P3] **Sijie Ruan**, Renchi Jiang, Song Tang, Yexin Li, Weixin Zhai, Xinhao Liu, Bingbing Hu, Hanning Yuan, Caicong Wu, Shuliang Wang. *Predictive Mobile Refueling for Agricultural Machinery via Deep Reinforcement Learning*. **KDD’26**.
- [P4] Zhenghong Wang*, **Sijie Ruan***, Tianqiang Huang*, Haoyi Zhou, Shanghang Zhang, Yi Wang, Leye Wang, Yu Liu. *A Lightweight Multi-Layer Perceptron for Efficient Multivariate Time Series Forecasting*. **KBS**, 2024.
- [P5] Yiqing Zou, Hanning Yuan, Qianyu Yang, Ziqiang Yuan, Shuliang Wang, **Sijie Ruan***. *Meta Dynamic Graph for Traffic Flow Prediction*. **AAAI’26**.
- [P6] Kaijun Liu, **Sijie Ruan**†, Liang Zhang, Cheng Long, Shuliang Wang, Liang Yu. *Generative Human Trajectory Recovery via Embedding-Space Conditional Diffusion*. **ICML’25**.
- [P7] Huaxu Han, Shuliang Wang, **Sijie Ruan**†, Qianyu Yang, Yuxuan Liang, Ziqiang Yuan, Cheng Long, Hanning Yuan, Yu Zheng. *AdaMove: Efficient Test-Time Adaptation for Human Mobility Prediction*. **ICDE’25**.
- [P8] **Sijie Ruan**, Jie Bao, Yuxuan Liang, Ruiyuan Li, Tianfu He, Chuishi Meng, Yanhua Li, Yingcai Wu, Yu Zheng. *Dynamic Public Resource Allocation Based on Human Mobility Prediction*. **UbiComp’20**.
- [P9] Shuliang Wang, Xinyu Pan, **Sijie Ruan**†, Haoyu Han, Ziyu Wang, Hanning Yuan, Jiabao Zhu, Qi Li. *DiffCrime: A Multimodal Conditional Diffusion Model for Crime Risk Map Inference*. **KDD’24**.
- [P10] Shuliang Wang, Ziyu Wang, **Sijie Ruan**†, Haoyu Han, Keqin Xiong, Hanning Yuan, Ziqiang Yuan, Guoqing Li, Jie Bao, Yu Zheng. *DelvMap: Completing Residential Roads in Maps Based on Couriers’ Trajectories and Satellite Imagery*. **TGRS**, 2024.
- [P11] Shuliang Wang, Jiabao Zhu, Kaibo Wang and Sijie Ruan†. *LeCDSR: Large Language Model Enhanced Cross-Domain Sequential Recommendation*. **Information Fusion**, 2026.
- [P12] Shuliang Wang, Xiaoting Leng, **Sijie Ruan***, Dingqi Yang, Yicheng Tang, Qianyu Yang, Qianxiong Xu, Jiabao Zhu, Hanning Yuan*. *Explicit Intent-Enhanced Knowledge Distillation for Trip Recommendation*. **AAAI’26**.
- [P13] **Sijie Ruan**, Yiqing Zou, Qianyu Yang, Haoyu Han, Yeting Zhang, Ziqiang Yuan, Hanning Yuan, Shuliang Wang. *Spatial Hierarchical Meta-Learning for Single-Point Map Matching*. **KDD’25**.
- [P14] Shuliang Wang, Jiabao Zhu, Yi Wang, Chen Ma†, Xin Zhao, Yansen Zhang, Ziqiang Yuan, **Sijie Ruan**†. *Hierarchical Gating Network for Cross-Domain Sequential Recommendation*. **TOIS**, 2025.
- [P15] Shuliang Wang, Qianyu Yang, **Sijie Ruan**†, Cheng Long, Ye Yuan, Qi Li, Ziqiang Yuan, Jie Bao, Yu Zheng. *Spatial Meta Learning with Comprehensive Prior Knowledge Injection for Service Time Prediction*. **TKDE**, 2025.
- [P16] **Sijie Ruan**, Cheng Long, Jie Bao, Chunyang Li, Zisheng Yu, Ruiyuan Li, Yuxuan Liang, Tianfu He, Yu Zheng. *Learning to Generate Maps from Trajectories*. **AAAI’20**.
- [P17] **Sijie Ruan**, Cheng Long, Xiaodu Yang, Tianfu He, Ruiyuan Li, Jie Bao, Yiheng Chen, Shengnan Wu, Jiangtao Cui, Yu Zheng. *Discovering Actual Delivery Locations from Mis-Annotated Couriers’ Trajectories*. **ICDE’22**.
- [P18] **Sijie Ruan**, Zi Xiong, Cheng Long, Yiheng Chen, Jie Bao, Tianfu He, Ruiyuan Li, Shengnan Wu, Zhongyuan Jiang, Yu Zheng. *Doing in One Go: Delivery Time Inference Based on Couriers’ Trajectories*. **KDD’20**.
- [P19] **Sijie Ruan**, Xi Fu, Cheng Long, Zi Xiong, Jie Bao, Ruiyuan Li, Yiheng Chen, Shengnan Wu, Yu Zheng. *Filling Delivery Time Automatically Based on Couriers’ Trajectories*. **TKDE**, 2023.
- [P20] **Sijie Ruan**, Cheng Long, Zhipeng Ma, Jie Bao, Tianfu He, Ruiyuan Li, Yiheng Chen, Shengnan Wu, Yu Zheng. *Service Time Prediction for Delivery Tasks via Spatial Meta-Learning*. **KDD’22**.
- [P21] **Sijie Ruan**, Keqin Xiong, Shuliang Wang, Jing Geng, Jie Bao, Yu Zheng. *A Survey of Urban Geographic Information Inference Driven by Crowdsourced Spatio-Temporal Data (in Chinese)*. **Acta Electronica Sinica**, 2023.
- [P22] Yue Hu, **Sijie Ruan**†, Yuting Ni, Huajun He, Jie Bao, Ruiyuan Li, Yu Zheng. *SALON: A Universal Stay Point-Based Location Analysis Platform (Demo Paper)*. **SIGSPATIAL’21 (Best Demo Paper Award)**.